

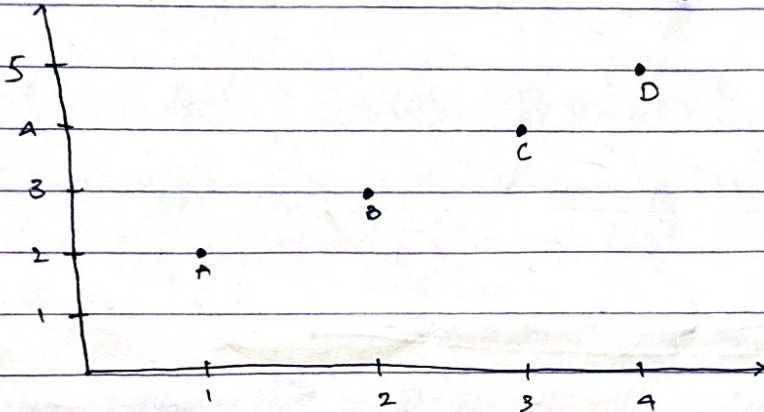
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COM232

CASINO CUSTOMER DATASET

| CUSTOMER | AVG BET PER VISIT (USD) | VISITS PER MONTH |
|----------|-------------------------|------------------|
| A        | 1                       | 2                |
| B        | 2                       | 3                |
| C        | 3                       | 4                |
| D        | 4                       | 5                |

1) PLOT THE DATASET ON A 2 DIMENSIONAL PLANE



2) COMPUTE THE MEAN OF EACH FEATURE

$$\mu_{x1} = \frac{1 + 2 + 3 + 4}{4} = \frac{10}{4} = 2.5 \quad \mu_{x2} = \frac{2 + 3 + 4 + 5}{4} = \frac{14}{4} = 3.5$$

$$\mu = \begin{bmatrix} 2.5 & 3.5 \end{bmatrix}$$

3A)

CENTER THE DATASET BY SUBTRACTING THE MEAN FROM EACH SAMPLE.

$$3) \quad A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix} \quad A_c = \begin{bmatrix} 1-2.5 & 2-3.5 \\ 2-2.5 & 3-3.5 \\ 3-2.5 & 4-3.5 \\ 4-2.5 & 5-3.5 \end{bmatrix} \quad 4) \quad A_c = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

5) COMPUTE THE SAMPLE COVARIANCE MATRIX OF THE CENTERED DATA

$$\hat{\Sigma} = \frac{1}{n-1} A_c^T A_c \quad A_c = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \quad A_c^T = \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix}$$

$$\hat{\Sigma} = \frac{1}{n-1} \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix} \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

$$\hat{\Sigma} = \frac{1}{3} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \quad \hat{\Sigma} = \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix}$$

6) COMPUTE THE EIGEN VALUES OF THE COVARIANCE MATRIX

$$\det(A - \lambda I) = 0$$

$$= \left| \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$= \begin{vmatrix} 1.667 - \lambda & 1.667 \\ 1.667 & 1.667 - \lambda \end{vmatrix} = 0$$

$$= (1.667 - \lambda)^2 - (1.667)^2 = 0 \quad a^2 - b^2 = (a-b)(a+b)$$

$$= [(1.667 - \lambda) - 1.667] \times [(1.667 - \lambda) + 1.667]$$

$$= 1.667 - \lambda - 1.667 = -\lambda$$

$$= 1.667 - \lambda + 1.667 = 3.334 - \lambda$$

$$= (-\lambda)(3.334 - \lambda) = 0$$

$$\boxed{\lambda_1 = 0, \lambda_2 = 3.334}$$

7) COMPUTE THE CORRESPONDING EIGENVECTORS

$$A - \lambda I = 0$$

$$A = \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix} \quad V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= \left( \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\lambda = 3.334$$

$$= \left( \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix} - \begin{bmatrix} 3.334 & 0 \\ 0 & 3.334 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$= \begin{bmatrix} -1.667 & 1.667 \\ 1.667 & -1.667 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$= \begin{bmatrix} -1.667v_1 + 1.667v_2 = 0 \\ 1.667v_1 - 1.667v_2 = 0 \end{bmatrix} \quad \begin{bmatrix} 1.667v_2 = 1.667v_1 \\ 1.667v_1 = 1.667v_2 \end{bmatrix} = \begin{bmatrix} v_2 = v_1 \\ v_1 = v_2 \end{bmatrix}$$

$$\boxed{V = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

f) NORMALIZE THE EIGENVECTOR

$$u = \frac{V}{\|V\|}$$

$$\|V\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\lambda = 3.334$$

$$u = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

a) CHECK IF THE EIGENVECTORS AND EIGENVALUES ARE CORRECT USING  $Av = \lambda v$

$$Av = \lambda v$$

$$= \begin{bmatrix} 1.667 & 1.667 \\ 1.667 & 1.667 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3.334 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.667 + 1.667 \\ 1.667 + 1.667 \end{bmatrix} = \begin{bmatrix} 3.334 \\ 3.334 \end{bmatrix}$$

$$= \begin{bmatrix} 3.334 \\ 3.334 \end{bmatrix} = \begin{bmatrix} 3.334 \\ 3.334 \end{bmatrix}$$

10) IDENTIFY THE FIRST PRINCIPAL COMPONENT, EXPLAIN WHY THIS COMPONENT IS CHOSEN OVER THE OTHERS.

- THE OTHER PCA VALUE IS 0, AND SINCE THE CHOSEN PCA WHICH IS 3.334 IS HIGHER THAN THE  $\lambda=0$  THEREFORE WE WILL USE THE HIGHER ONE BECAUSE IT HAS THE MAXIMUM VARIANCE.

11) PROJECT THE CENTERED DATA ONTO THE FIRST PRINCIPAL COMPONENT.

$$\text{NEW PROJECTION} = A_c \times u$$

$$= \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \times \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

$$= \frac{-1.5 - 1.5}{\sqrt{2}} = \frac{-3}{\sqrt{2}} \quad \frac{0.5 + 0.5}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\frac{-0.5 - 0.5}{\sqrt{2}} = \frac{-1}{\sqrt{2}} \quad \frac{1.5 + 1.5}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

12) RESULTING 1-D REPRESENTATION

